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Deforestation-induced emissions from mining energy transition minerals

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Yifei Quan [1] & Jie-Sheng Tan-Soo [1,2]

The global transition to low-carbon energy depends on energy transition minerals (ETMs). Yet, the extraction of these minerals often occurs in biodiverse and carbon-rich forests, potentially undermining their climate benefits. Here we provide global, causally identified estimates of deforestation and related GHG emissions attributable to ETM mining, combining nearly 3,000 projects with satellite-based forest-change data. Using a staggered difference-in-differences design, we find that ETM mining causes sustained forest loss—averaging ~20% within 10-km buffers over 15 years—comparable in magnitude to traditional minerals such as coal and gold. These losses are disproportionately concentrated in tropical forests with high climate mitigation potential. Incorporating deforestation-related emissions increases the mining-stage carbon footprint of ETMs by 63% on average and up to 98% for certain minerals. Our findings reveal mining-induced land-use change as a major but overlooked source of emissions in global energy transition.

Given the outsized contribution of fossil fuels towards GHG emissions, successful climate mitigation requires rapid clean energy transition. Energy transition minerals (ETMs) are essential to this shift, serving as key inputs for technologies such as electric vehicles and solar panels. According to the International Energy Agency, meeting the net-zero emissions scenario will require global ETM demand to quadruple by 2040[1]. Consequently, ETM mining is expanding sharply, with projects rising by 50% over the past decade[2].

Mineral extraction has historically required extensive land clearing[3], resulting in substantial deforestation[4–6]. As forest ecosystems function as carbon sinks[7], ETM extraction creates a paradox where their decarbonization objective induces GHG emissions through deforestation.

However, mining-induced deforestation and associated emissions are rarely reported by mining companies or considered in life-cycle assessments[8,9]. For example, a recent study documented that land transformation factors for nickel mining have been greatly underestimated[10]. Still, it remains uncertain how much of GHG emissions stem specifically from ETM-related deforestation or how these previously overlooked emissions alter the carbon footprint of ETMs.

Geographically, mining projects—both for ETMs and traditional minerals—frequently intersect with densely forested regions (Fig. 1). Despite similar geographic contexts, research on mining-driven deforestation mainly focuses on traditional minerals, especially coal and gold[6,11,12]. Direct evidence linking ETM mining to deforestation remains scarce. Additionally, ETM projects largely overlap with lands of Indigenous peoples and local communities (IPLCs)[13,14], introducing local governance dimensions into the ETM–deforestation link.

In response to increasing ETM mines, companies and public institutions have launched new initiatives[15], such as the Sustainable Critical Minerals Alliance at COP15. However, their effectiveness in ensuring environmentally responsible mining, particularly compared with traditional minerals, remains unclear.

Most existing studies on mining-driven deforestation rely on correlative analysis that identify time-series changes in forest cover[11,16–18]. While illuminating, such approaches are susceptible to confounding from other land-use pressures such as agricultural expansion near mine sites[19]. More recent studies use causal inference designs that compare mined areas with otherwise similar unmined sites to isolate the impact of mining. Findings, however, are mixed across regions[4,19–21], plausibly because national/subnational studies embed

[1]Lee Kuan Yew School of Public Policy, National University of Singapore, Singapore, Singapore. [2]Institute for Environment and Sustainability, National University of Singapore, Singapore, Singapore. e-mail: jiesheng.tan@nus.edu.sg

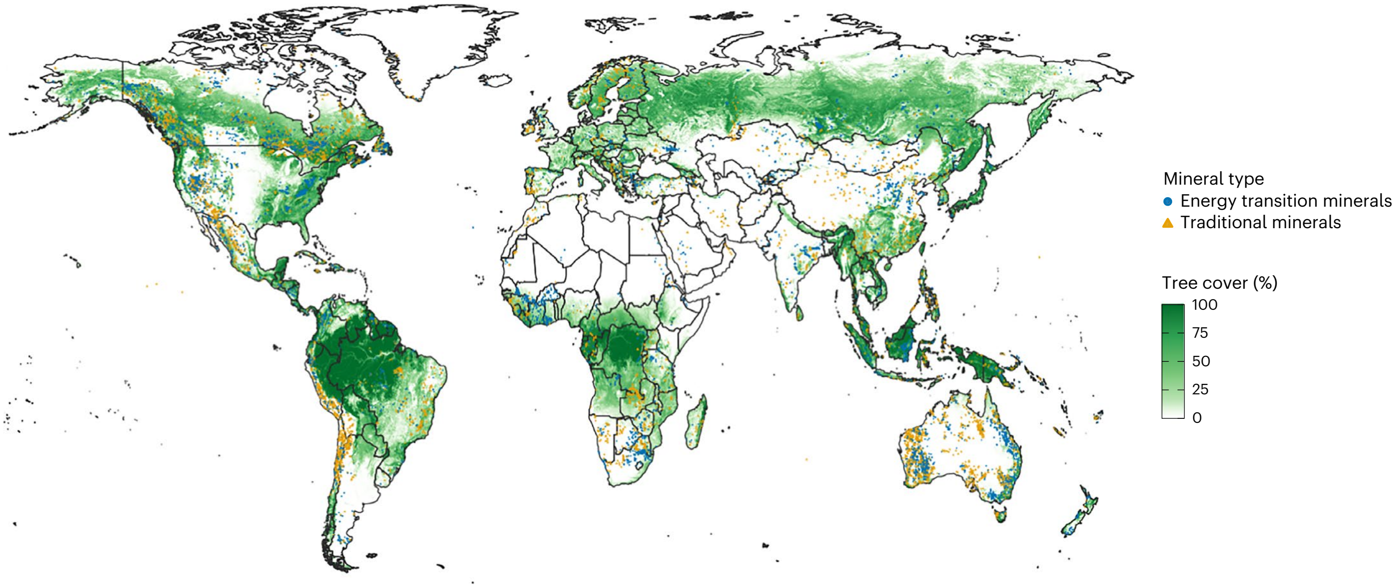

Fig. 1 | Global overlap between forests and mines. Blue and yellow points mark mining projects for ETMs and traditional minerals, respectively (n = 12,538). Tree cover is derived from satellite-based canopy coverage for the year 2000. Basemap from Natural Earth (https://www.naturalearthdata.com/).

country-specific governance, development and land-tenure differences. This, in turn, warrants a globally comparable analysis accounting for cross-country heterogeneity.

We fill this gap with a global, data-driven evaluation of deforestation attributable to ETM mining. Leveraging comprehensive project data and high-resolution forest-change maps, we use a difference-in-differences (DID) design to estimate the impacts of mining on surrounding forests. We compare these effects across ETM and traditional projects to assess whether recent sustainability pledges have triggered reduced environmental harms. Further, we examine cross-country variation in forest loss, emphasizing regions with high climate mitigation potential and explore how overlaps between ETM mines and IPLC lands affect deforestation outcomes. Finally, we quantify deforestation-driven GHG emissions to update the carbon footprints of ETMs.

Forest loss driven by ETM mining

We begin by estimating direct and indirect forest losses from ETM mining. Following existing specifications[11,20,22], we define 10-km and 50-km buffers around each mine to, respectively, capture direct and indirect impacts. The start of project construction serves as the event-study benchmark, allowing comparison of forest dynamics across early exploration stage versus subsequent construction and operational phases.

In our main analysis, control units are early-stage inactive projects abandoned before construction. These sites share geological characteristics with treated mines but experience limited mining disturbance, allowing us to isolate the causal effect of mining on deforestation.

Pooling all minerals, the full-sample estimate demonstrates significant direct forest loss, declining by ~20% over 15 years after construction (Supplementary Fig. 1). Pretreatment coefficients are small and statistically insignificant, consistent with parallel trends. To examine temporal dynamics, we replace cumulative forest extent with annual loss and see that deforestation intensifies during the first 3 years of construction, before diminishing and becoming statistically insignificant after 10 years (Supplementary Fig. 2).

Comparing ETMs with traditional minerals, we observe similar trajectories (Fig. 2). Annual deforestation rates are 1.38% for ETMs and 1.44% for traditional minerals (20.7% versus 21.6% over 15 years). Thus, despite recent sustainable initiatives, ETM projects cause forest loss comparable to traditional mining, even in recent cohorts (Supplementary Fig. 3).

We next examine indirect deforestation effects from expansion activities around mine sites[23]. We detect no significant deforestation between 10 km and 50 km (Supplementary Fig. 4). Nevertheless, in previously identified deforestation hotspots (for example, Indonesia and Brazil)[4,5], impacts persist up to 60 km (Supplementary Fig. 5), highlighting regional heterogeneity in land-use impacts.

Several robustness checks support our main findings. These include varying buffer ranges[3,24], using alternative DID estimators[25–27] and modifying covariates and control-group definitions (Supplementary Methods and Supplementary Figs. 3, 6 and 7).

Cross-country variation in forest loss

Our global analysis covers 32 countries, spanning tropical to boreal zones and both developed and developing economies. To capture heterogeneity, we estimate deforestation effects separately by country.

Country-level annual forest loss rates range greatly from 0.15% in Mexico to 4.29% in the DRC (Fig. 3). Notably, deforestation is most pronounced where forests constitute a high share of land, rather than where total forest area is extensive.

We further correlate country-level deforestation estimates with the climate mitigation potential of forests to account for differing climate mitigation capabilities across forest types. Using the average carbon accumulation rate over the next 30 years as a proxy[28], we find positive correlation (ρ = 0.58) between deforestation rates and mitigation potential. Highest rates occur in the DRC, Gabon and Indonesia—regions with the greatest mitigation potential, implying disproportionate losses in the most climate-valuable forests.

Deforestation also strongly correlates with economic status and governance quality of countries (Supplementary Fig. 6). This heterogeneity links mining management performance to broader development disparities and indicates where conservation investments may be most urgent.

Intersection between ETM mining and IPLC lands

ETM mining extensively overlaps IPLC-held lands[14]. Because IPLC territories often coincide with ecologically important areas[13,29], we compare deforestation effects for projects intersecting IPLC versus non-IPLC lands. Using a global map of IPLC lands[30], we intersect mining buffers with IPLC polygons (Supplementary Methods). Of 812 ETM projects analysed, 48% overlap IPLC lands.

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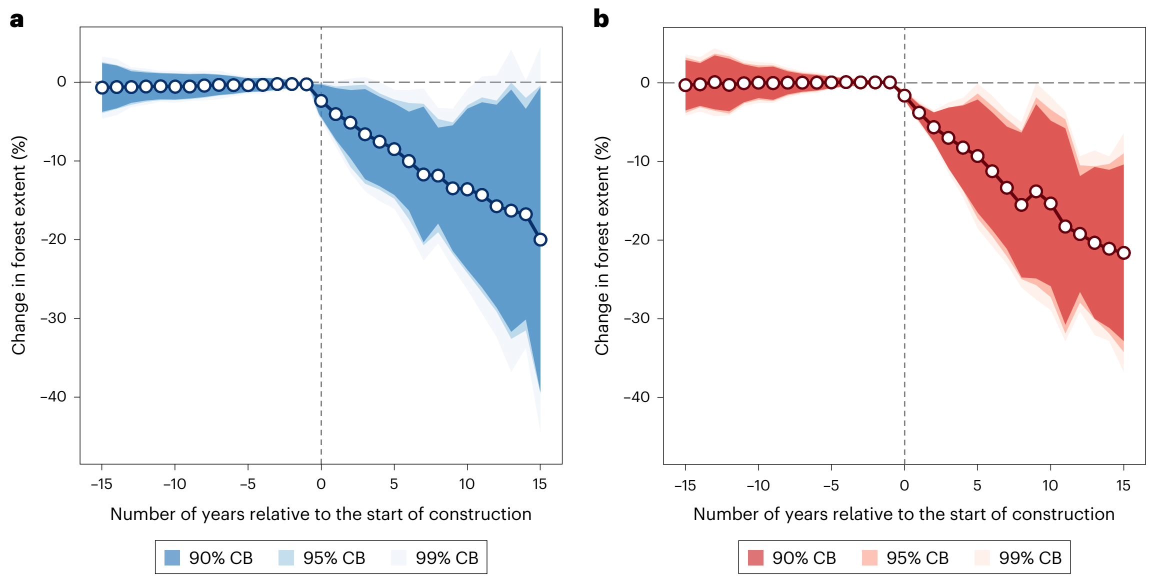

Fig. 2 | Dynamic effects of mining activity on forest extent. a,**b**, Estimated direct changes in forest extent within 10-km buffer around ETM mining projects (**a**) and traditional mineral mining projects (**b**), relative to the start year of construction. Points are group-time average treatment effects; bootstrapped standard errors are clustered at the country level; shaded areas denote simultaneous confidence bands (CB) at varying significance levels. Estimates are computed from mine sites with ≥30% forest cover in 2000.

Our first comparison between IPLC and non-IPLC lands reveals negligible differences in forest loss (1.37% versus 1.48%; Supplementary Fig. 9). Motivated by commons governance theory[31–33], we further split IPLC lands by legal recognition: acknowledged lands are those legally recognized as collective property or officially allocated, while unacknowledged lands have no formal recognition[30]. Of ETM projects, 83% are located on acknowledged lands and 17% are on unacknowledged lands.

The contrast between these groups is stark (Fig. 4a). On acknowledged IPLC lands, mining results in moderate forest loss of 5.9% within 10 years, after which impacts taper off. On unacknowledged lands, deforestation is severe and persistent, reaching 27.2% over 10 years. This heterogeneity highlights the importance of formally recognized land rights in sustainable resource management.

Globally, recognition of IPLC rights is polarized (Fig. 4b). Where recognition is weak (for example, Gabon, the DRC and Indonesia), forest loss is highest and conflict rises with economic activity falls (Supplementary Table 1). These patterns situate environmental governance within a broader development context.

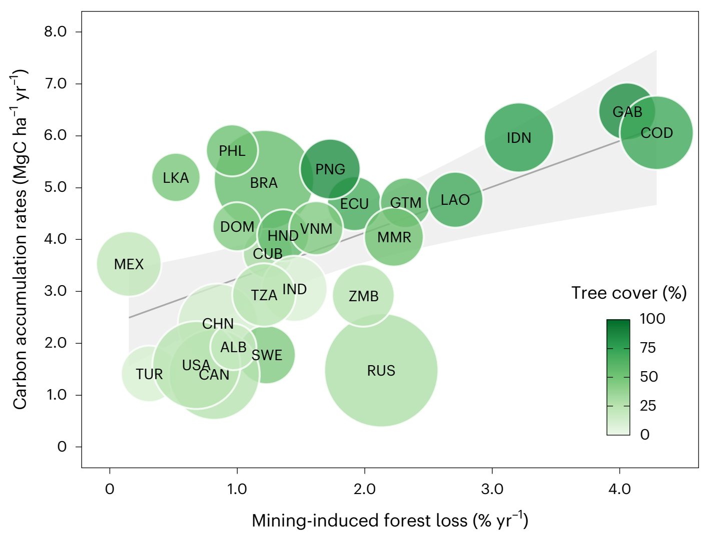

Fig. 3 | Country-level deforestation from ETM mining and its correlation with forest mitigation potential. The horizontal axis represents country-specific mining-induced annual forest loss from event-study estimates. The vertical axis represents country-level carbon accumulation rates from 2020 to 2050, used as a proxy for climate mitigation potential. Each bubble corresponds to a country ($n = 27$); country labels correspond to ISO 3166-1 alpha-3 codes; bubble size reflects total forest area in 2000 and bubble colour indicates tree cover density in 2000. Five countries (Australia, Fiji, Finland, Germany and Madagascar) are excluded because of violation of the parallel trends assumption or insignificant post-treatment estimates. A linear fit is shown, where the fitted line represents the centre of shaded 95% confidence interval. The correlation coefficient is 0.58.

Climate implications of mining-induced deforestation

To quantify climate impacts of mining-induced deforestation, we rerun the DID model with GHG emissions as the outcome (Methods). On average, each ETM project yields about 100,400 tCO_2e) yr^{-1} (Supplementary Fig. 10), or 3.2 tCO_2e ha^{-1} within the 10-km buffer. Anticipating preconstruction forest clearing[34], we re-code treatment as a 1-, 2- and 3-year lead to capture overall deforestation-related emissions. The precision of our estimates is unchanged (Supplementary Table 2).

We further contextualize deforestation-induced emissions by deriving additional carbon footprint per mineral (Methods). To reflect spatial heterogeneity, we compute emission factors by ecozone and land tenure. Across nine ETMs, average emission factors range from 0.14 tCO_2e per t for chromium to 62.1 tCO_2e per t for silver (Table 1 and Supplementary Tables 3 and 4). These estimates are conservative lower bounds, as they exclude indirect climatic effects (for example, atmospheric CO_2 concentration, nutrient cycling and disaster risk)[35–37], which would further diminish the perceived climate benefits of ETMs.

Compared against current values from life-cycle inventory database[38], our calculations reveal that deforestation-driven emissions constitute 37%–98% of current mining-stage carbon footprints and 5%–32% of cradle-to-gate footprints. By adding this new component, we update ETM emission factors (Supplementary Table 5), which can be readily applied in mining-sector climate reporting and scope 3 accounting[39–41].

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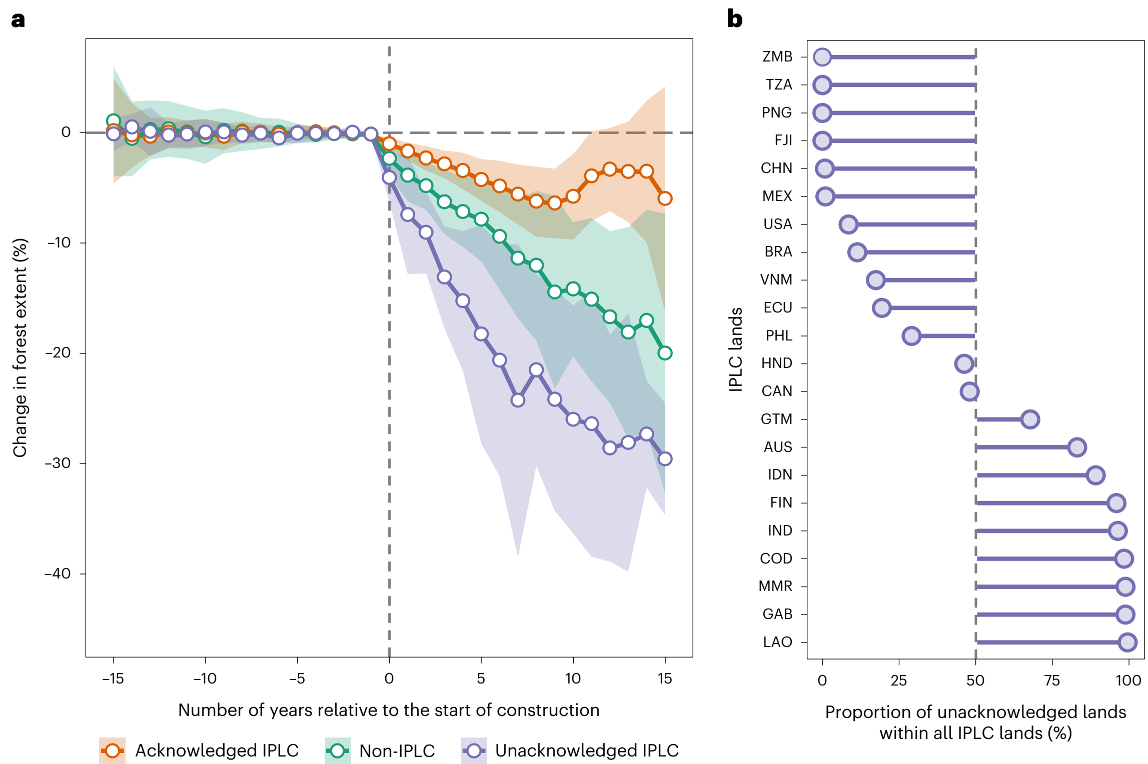

Fig. 4 | Heterogeneous forest loss from ETM mining across IPLC and non-IPLC lands. a, Estimated changes in forest extent surrounding ETM projects that overlap with acknowledged IPLC lands (orange), unacknowledged IPLC lands (purple) and non-IPLC lands (green), relative to the start year of project construction. Points represent group-time average treatment effects; shaded areas indicate 95% bootstrapped confidence intervals. **b**, Proportion of unacknowledged IPLC lands within all IPLC areas of a country. Estimates are computed from mine sites with ≥30% forest cover in 2000.

Table 1 | Emission factors of mining-related deforestation for ETMs by ecozone and land tenure

ETMs	Emission factor (tCO_2e per t of mineral)						
	Average[a]	Tropical[b]	Subtropical[c]	Temperate[d]	Boreal[e]	IPLC[f]	Non-IPLC[g]
Chromium	0.14	0.23	0.11	N/A	0.06	0.07	0.16
Cobalt	0.51	0.81	0.23	N/A	N/A	0.22	0.55
Copper	0.87	2.54	0.62	0.43	0.21	0.38	0.95
Graphite	0.14	0.21	0.16	0.09	N/A	0.06	0.15
Lithium	0.35	0.51	0.28	N/A	0.17	0.15	0.39
Manganese	0.23	0.32	0.16	N/A	N/A	0.09	0.27
Nickel	1.87	2.46	1.59	0.42	0.50	0.81	2.06
Silver	62.10	73.14	37.35	45.47	31.96	27.45	64.50
Zinc	0.53	0.81	0.66	0.28	N/A	0.22	0.65

Each row represents an ETM. [a]The average emission factor estimates. [b–e]Ecozone-specific emission factors based on the location of projects in tropical, subtropical, temperate and boreal forests, respectively. [f,g]Emission factors for projects located on acknowledged IPLC lands and non-IPLC lands, respectively. Estimates are computed from mine sites with ≥30% forest cover in 2000. N/A, no observations in the corresponding category. 95% confidence intervals are presented in Supplementary Tables 3 and 4.

Discussion

Global demand for ETMs is projected to rise sharply to support energy transitions[1]. Yet research on environmental impacts of ETM mining lags behind, with most previous work focused on traditional minerals[6,11,12,42]. Although public institutions have acknowledged the double-edged influence of ETMs and inked agreements promoting responsible mining, it remains unclear whether these efforts have mitigated environmental harm.

Using a global dataset of mining projects, we show that ETM mining drives substantial deforestation, with forest extent declining by an average of 20% within 15 years of operation. This highlights a fundamental trade-off between sustainable development goals SDG13 (climate action) and SDG15 (life on land). Notably, deforestation linked to ETMs is quantifiably comparable to that from traditional minerals, suggesting that growing sustainability commitments have yet to yield measurable improvements.

Our cross-country analysis uncovers pronounced heterogeneity: deforestation is greatest in regions with the highest mitigation potential, undermining opportunities for nature-based climate solutions. Moreover, national differences in deforestation correlate closely with economic and governance capacity, suggesting that developmental inequalities exacerbate environmental injustice.

The intersection of ETM mining and IPLC lands introduces an additional governance dimension. We find significantly lower deforestation on IPLC lands formally recognized by authorities, highlighting the importance of secure property rights. Yet, despite international endorsement of IPLC land tenure, 42% of such lands remain unrecognized globally[29].

Interestingly, nations with modest economic capacity but stronger IPLC rights, such as Tanzania and Zambia, exhibit moderate deforestation. This suggests that formalizing IPLC land tenure and enabling community governance could be an effective strategy even in resource-constrained contexts. In practice, one increasingly adopted mechanism is impact benefit agreements that legally bind arrangement between mining companies, governments and IPLCs[43–45]. These agreements can help to secure social license, distribute benefits more equitably and mitigate adverse environmental and social impacts[46,47].

Finally, we show that GHG emissions from mining-induced deforestation are substantial for ETMs, corresponding to 37%–98% of current mining-stage and 5%–32% of cradle-to-gate emissions across minerals. These figures probably represent lower bounds, as they exclude broader implications such as forest ecosystem services[37] and biophysical process alterations[36]. Thus, the true environmental footprint of ETMs is systematically underestimated.

Overall, our findings underscore the urgent need for comprehensive accounting of ETM-related environmental impacts within mining-sector sustainability disclosures and land-use GHG inventories[8,39]. Emerging supply-chain regulations already compel businesses to incorporate suppliers' carbon footprint into their procurement decisions. To aid in this effort, we provide updated emission factors to improve the accuracy of GHG accounting.

Despite these contributions, several limitations remain. First, satellite-based datasets may not fully capture forest regeneration after initial clearing or offsets such as off-site tree planting[48,49]. Accordingly, our estimates reflect gross losses relative to baseline cover rather than net change. Second, our results are applicable to sites with ≥30% tree cover in 2000 and should not be extrapolated to inherently non-forested settings (for example, arid ecozones). Accordingly, cross-country and IPLC comparisons should also be interpreted within this forest-cover threshold. Third, IPLC boundaries derive from an ongoing programme and may omit some territories, potentially conflating unrecorded lands with non-IPLC areas.

From a general-equilibrium perspective, mining booms can shift land-use pressures via various indirect channels, for example, labour reallocation from agricultural sector, infrastructure expansion, in-migration and higher demand for food and timber, making the net landscape-wide effect on deforestation ambiguous. Evidence on these reallocation channels remains limited, making this an important avenue for future research.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-025-02520-w.

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Methods

Mining project data

We compile global mining project data from S&P Capital IQ Pro[2], one of the most comprehensive project-level databases[10,14]. As of April 2025, it contains 37,886 projects across all development stages, from exploration to closure.

We apply a series of filtering steps to construct the analytic sample while preserving representativeness (Supplementary Fig. 11). For valid entries, we generate 10-km and 50-km buffers around project coordinates to delineate zones of direct and indirect environmental impact, following empirical observations[23] and standards in the literature[11,22].

Furthermore, we retrieve clear development timelines for each project. In addition to temporal variables from S&P, we manually supplement missing information using historical logs and analyst commentaries. While production start dates are generally reported, construction dates are sometimes absent; for such cases, we impute them using average lead times from comparable projects.

Projects are categorized as either ETM or traditional mineral based on previous publications[14,50–54]. Our primary list includes 14 minerals identified as ETMs across all six sources. For robustness, we use an expanded list of 28 ETMs covered in at least one source (Supplementary Fig. 3a). Traditional minerals include those in the S&P database but absent from these lists, primarily coal, gold and diamonds (Supplementary Table 6).

A categorization challenge arises with co-producing projects (for example, gold and copper mines), which comprise 18% of the sample. In these cases, we classify by the primary commodity listed in S&P. Excluding such mixed projects does not alter results (Supplementary Fig. 3b).

Forest-change and emissions data

We obtain forest loss data for 2000–2023 from the satellite-based Global Forest Watch dataset[55] with ~30-m resolution. We use two layers: 'treecover2000' (baseline canopy coverage in 2000) and 'lossyear' (subsequent year of forest loss). Following conventions, we define 'forest' as pixels with canopy cover ≥30% in 2000 and restrict the sample to mining buffers where at least 30% of pixels meet this threshold[20]. Annual forest extent is quantified by counting remaining pixels within each buffer for every year and directly converting pixel counts into definitive area estimates is inappropriate. Forest loss is assumed irreversible over the mining life cycle.

GHG emissions associated with forest loss are extracted from a global carbon flux map[56]. As each pixel in this dataset corresponds spatially with the 'lossyear' layer, emissions data can be accurately matched to the year of deforestation. Pixel-level emissions are aggregated annually within each buffer to yield total GHG emissions from mining-induced deforestation.

DID model

We estimate the causal impact of mining on forest loss using a DID model that compares treated and control projects over time. Fundamentally, a DID model is to mimic a controlled experiment by observing differences in changes between two groups that are similar except for the evaluated treatment.

The key assumption of DID is parallel trends—the treated and control projects would have followed similar trajectories in forest change in the absence of mining. Any divergence after mining begins can thus be attributed to mining activity itself. Compared with correlative analysis, this design mitigates biases from confounders that affect both the likelihood of mining and deforestation (for example, geological characteristics) as long as they affect both the treated and control units equally.

Reliable DID design is contingent on comparability between treatment and control groups. Our treated group consists of active projects either under construction or in production. We exclude early-stage projects (for example, exploration) due to minimal environmental disturbance and closed projects to avoid reclamation effects. Control group includes inactive projects halted or abandoned at early stages, which resemble treated projects across many characteristics (Supplementary Table 7) but did not proceed as a result of economic or geological constraints. As robustness checks, we also use early-stage active projects as an alternative (Supplementary Fig. 3e).

Our analytic dataset covers 2000–2023 and include 2,736 projects (819 treated; 1,917 controls; $n = 65{,}664$). We conduct a series of statistical checks to verify representativeness of our sample (Supplementary Fig. 12). Since treatment timing varies across projects, classic two-way fixed-effects estimator could be biased. We hence apply an advanced DID estimator designed to address varying treatment timing and treatment effect heterogeneity[57]. We estimate event-study effects using a universal baseline to avoid asymmetric differencing[58] and coefficients are reported relative to the final preconstruction year. All data processing and covariate definitions are detailed in Supplementary Methods.

Carbon footprint of mining-related deforestation

We further quantify additional carbon emissions from ETM-induced deforestation and convert them into incremental carbon footprints by mineral.

First, ETM projects are grouped by their primary mineral and their average annual deforestation-related emissions are estimated using the DID. Then, we calculate emission factors (tCO_2e per t of mineral) by dividing emissions by production capacity (as a proxy for actual output) reported in the S&P database.

Next, we compare these values against emission estimates from the Ecoinvent database. For each mineral, we collect emissions factors for the mining-stage and cradle-to-gate life cycle. Notably, emissions from land-use change are typically excluded from these values[38].

To illustrate, consider nickel mining. According to the DID estimates, an average nickel project generates ~48,200 tCO_2e annually from deforestation, with an average output of 25,800 t of nickel per year. This corresponds to a deforestation-related carbon footprint of 1.9 tCO_2e per t. By comparison, Ecoinvent reports mining-stage and cradle-to-gate emissions of 4.6 $tCO_2e\ t^{-1}$ and 7.6 $tCO_2e\ t^{-1}$, respectively. Therefore, excluding deforestation underestimates the mining-stage emissions of nickel by 41% and its cradle-to-gate emissions by 25%. Full calculation details are provided in Supplementary Tables 3–5.

Data availability

The S&P Capital IQ Pro database used in this study is proprietary and available only to licensed subscribers. Forest change and GHG emission datasets are publicly available from the Global Forest Watch (http://data.globalforestwatch.org/). Supplementary Methods provide details on access to auxiliary data sources. The anonymized dataset is available via Zenodo at https://doi.org/10.5281/zenodo.17251281 (ref. 59). Source data are provided with this paper.

Code availability

All analyses were conducted in R (v.4.5.0) and the corresponding code scripts are available via Zenodo at https://doi.org/10.5281/zenodo.17251281 (ref. 59).

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Author contributions

Y.Q. and J.-S.T.-S. conceived the study and designed the research. Y.Q. performed the analysis with support from J.-S.T.-S. Resources were provided by J.-S.T.-S. All versions of the paper were written by Y.Q. and J.-S.T.-S.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Jie-Sheng Tan-Soo.

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